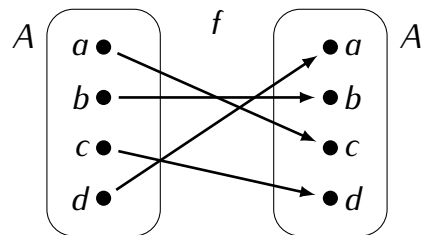


Definition 10.1

A *permutation* of a set A is a function $f: A \rightarrow A$ which is a bijection.

**Definition 10.2**

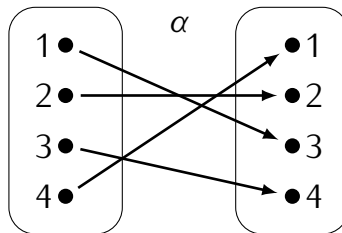
Let A be a set. The *permutation group* of A is a group $S(A)$ defined as follows:

- **Elements of $S(A)$:** permutations $f: A \rightarrow A$.
- **Group operation:** composition of functions $g \circ f$.
- **The identity element:** the function $\varepsilon: A \rightarrow A$, $\varepsilon(x) = x$ for all $x \in A$.
- **The inverse of f :** the inverse permutation f^{-1} .

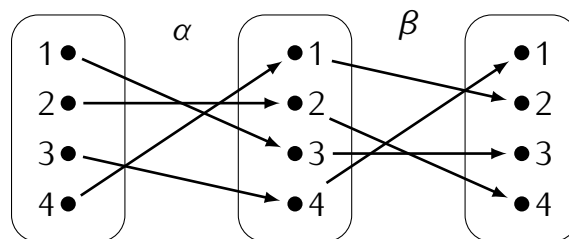
Definition 10.3

For $n \geq 1$ the group S_n is the group of permutations of the set $A = \{1, 2, \dots, n\}$. This group is called the *symmetric group on n letters*.

Matrix notation of permutations:



Composition:



Theorem 10.4

For any $n \geq 1$ we have $|S_n| = n!$

Dihedral groups and permutation groups

Note. The groups S_n are non-abelian for $n > 2$